

Probing Deep into Temporal Profile Makes the Infrared Small Target Detector Much Better

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Abstract—Infrared small target (IRST) detection is challenging in simultaneously achieving precise, robust, and efficient performance due to extremely dim targets and strong interference. Current learning-based methods attempt to leverage “more” information from both the spatial and the short-term temporal domains, but suffer from unreliable performance under complex conditions while incurring computational redundancy. In this paper, we explore the “more essential” information from a more crucial domain for the detection. Through theoretical analysis, we reveal that the global temporal saliency and correlation information in the temporal profile demonstrate significant superiority in distinguishing target signals from other signals. To investigate whether such superiority is preferentially leveraged by well-trained networks, we built the first prediction attribution tool in this field and verified the importance of the temporal profile information. Inspired by the above conclusions, we remodel the IRST detection task as a one-dimensional signal anomaly detection task, and propose an efficient deep temporal probe network (DeepPro) that only performs calculations in the time dimension for IRST detection. We conducted extensive experiments to fully validate the effectiveness of our method. The experimental results are exciting, as our DeepPro outperforms existing state-of-the-art IRST detection methods on widely-used benchmarks with extremely high efficiency, and achieves a significant improvement on dim targets and in complex scenarios. We provide a new modeling domain, a new insight, a new method, and a new performance, which can promote the development of IRST detection. Codes are available at <https://tinalrj.github.io/DeepPro/>.

Index Terms—Infrared small target detection, temporal profile, anomaly detection, remote sensing object detection, temporal probes, attribution analysis.

1 INTRODUCTION

The objective of Infrared Small Target (IRST) detection [1], [2] is to identify and localize moving targets that are typically “small” and “dim”¹ from infrared images. It has been a longstanding, fundamental yet challenging problem in infrared remote sensing [3], [4], and has broad applicability in both civilian and military domains, including reconnaissance, precision guidance [5], [6], maritime surveillance [7], [8], early fire warnings [9], [10], vehicle detection [11], [12], and UAV monitoring [13], [14]. In recent years, IRST detection has attracted increasing attention and made measurable progress over the past few decades; however, most IRST detection methods have not been capable of simultaneously achieving outstanding performance in *precision*, *robustness*, and *efficiency* due to the following challenges [10], [15], [16], [17].

- *High-precision related challenges*. On the one hand, the targets themselves are usually very small (e.g., typically occupy $\leq 9 \times 9$ pixels), dim (e.g., with an $\text{SNR} \leq 3$) [6], [18], [19], and with insufficient appearance features, such as shape and texture information. Such characteristics increase the difficulty of developing effective feature representations. On the other hand, the background is usually nonsmooth and nonuniform, containing clutter and noise. Clutter refers to imaged real things, while

noise refers to electronic noise and inaccuracies of the signal processor. They almost occupy the entire image, and some of them are strong and dynamic, similar to the targets as shown in Fig. 1(a) and (b). This complicates the effective suppression of false alarms.

- *High-robustness related challenges*. Apart from the aforementioned challenges, the target characteristics evolve temporally under motion conditions. The intensity of the target varies, and the local background neighborhood of the target also changes dynamically. Furthermore, practical scenarios involve diverse complex backgrounds with distinct clutter properties and various targets with different grayscale distributions, as shown in Fig. 1(a) and (b). These complex factors make it difficult to detect targets continuously and stably.
- *High-efficiency related challenges*. The application of IRST detection requires real-time processing of massive data streams, while IRST is extremely sparse in the spatial domain. It is difficult to balance sufficient feature extraction with efficient computation.

Therefore, in diverse complex scenarios, *which modeling domain can better exhibit the targets? Which prior information can more effectively capture the essential differences between target and background?* Exploring these questions is critical to achieve precise, robust, and efficient detection.

Existing deep learning-based methods focus on extracting “more” information to improve the detection performance. For example, many single-frame methods [18], [20], [21] extensively explore the local saliency and global information in the spatial domain via the complex network structure. Besides, some multi-frame methods [22], [23], [24], [25] explore motion information in the short-term spatial-temporal domain. Although performance improvements are achieved in some scenarios, these methods suffer from severe performance degradation under complex conditions. This comes from the limitation that the target

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1. “Small” means the target usually occupies a small portion of the overall image, and “dim” indicates that the target exhibits a weak signal intensity and low contrast relative to its background.

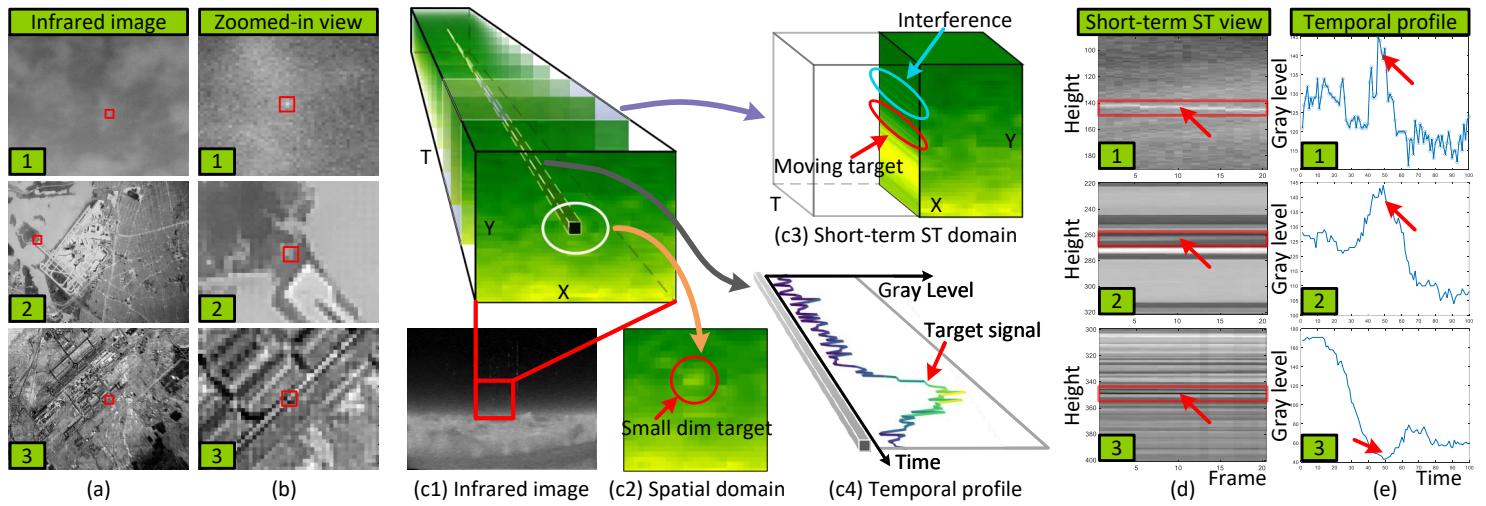


Fig. 1. Visualizations of IRSTs in different scenarios and typical detection difficulties in different domains. (a, b) Distinct targets in different scenarios (Cases 1, 2, and 3): in appearance, targets and clutters show small inter-class differences, while different targets exhibit marked intra-class variations. (c1) Infrared image. (c2) Spatial domain: small dim targets are barely observable. (c3) Short-term spatial-temporal (ST) domain: targets are indistinguishable from interferences. (c4) Temporal profile: records dynamic statistical details of all signals at fixed spatial location, where small dim target signals are complete and more prominent. The temporal profile contains more essential information. (d, e) More contrasts of the three cases. For clarity, targets are marked with red annotations. Corresponding dynamic demos are available at <https://tinalrj.github.io/DeepPro/>.

information is weak and that other discriminative information is insufficient in both domains, as illustrated in Fig. 1(c2) and (c3). In addition, extracting “more” information inevitably expands receptive fields, causing severe computational redundancy that fundamentally undermines real-time processing capabilities. Therefore, given our previous analysis, the focus of this work is to explore the “more essential” information from a more crucial profile and develop a highly effective and efficient method for IRST detection, especially under complex conditions.

In this paper, we study the inherent characteristics of IRST detection task in depth. We reveal that the temporal profile is crucial especially under complex conditions, which dramatically amplifies target saliency as shown in Fig. 1(c4) and (e). Through further theoretical analysis, we find that the temporal profile contains global temporal saliency information of target signals and correlation information between different signals, which are more essential for IRST detection. To validate the importance of the temporal profile in well-trained networks, we built the first prediction attribution [26], [27] tool that can analyze which input pixels impact the predictions of IRST detection networks. The attribution results show that the most influential pixels are concentrated in a small area centered on the target and continuously distributed along the time axis, which illustrates the importance of the temporal profile.

Inspired by the above research, we remodel the IRST detection task as a one-dimensional signal anomaly detection task in the temporal profile, where targets manifest as statistical outliers in sequential signals. Then, we design a pixel-wise temporal probe mechanism (TPro) to extract essential information in the temporal profile. Based on TPro, we propose an effective and efficient deep temporal probe network (i.e., DeepPro) for IRST detection, which performs multiplications and additions only in the time dimension. We conducted numerous experiments on widely-used benchmarks. As shown in Fig. 2, our DeepPro achieves significantly superior performance as compared to existing state-of-the-art methods employing complex spatial-temporal computations. Meanwhile, our DeepPro exhibits higher robustness on dim targets and in complex scenarios with strong noise, achieving an 87.6% reduction in parameter count and a 13-FPS inference gain on 256×256 images compared to the

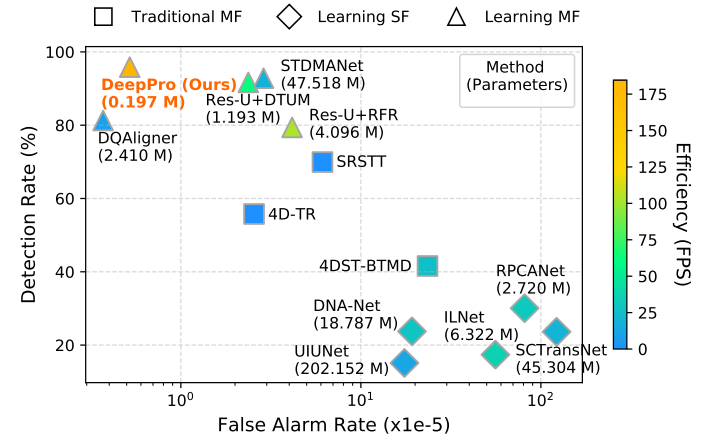


Fig. 2. Comparison of the detection performance, computational efficiency (Frames Per Second, FPS), and parameter counts of different deep learning-based methods on dim targets ($SNR \leq 3$) in the NUDT-MIRSST dataset. The color of each symbol represents the computational efficiency of the method. The yellow the color, the higher efficiency the method holds. Our proposed method is highly efficient and effective compared with other methods, including traditional multi-frame (MF) methods (SRSTT [7], 4DST-BTMD [28], and 4D-TR [29]), deep learning-based single-frame (SF) methods (DNA-Net [30], UIUNet [31], SCTransNet [20], RPCANet [32], and ILNet [116]), and deep learning-based multi-frame methods (Res-UNet+DTUM [6], STDManNet [23], Res-UNet+RFR [4], and DQAligner [117]).

existing lightest and second-fastest single-frame IRST (SIRST) detection network (i.e., ACM [46]).

The main contributions of this work are as follows.

- We reveal some new insights from a more crucial profile (i.e., temporal profile): long-term temporal information is much more essential for IRST detection, which includes global temporal saliency of target signals and correlation information between different signals. We validated the importance of the temporal profile in IRST detection, by developing the first predictive attribution tool.
- Inspired by our research, we remodel the IRST detection task as a one-dimensional signal anomaly detection task. Then, we propose DeepPro to leverage the essential tem-

poral profile information with multiply-add operations only in the time dimension.

- Experimental results show that DeepPro achieves a significant performance improvement in IRST detection with extremely high efficiency, and has high robustness to dim targets and scenes with strong noise.

2 RELATED WORK

In this section, we briefly review the major works in SIRST detection, multi-frame IRST (MIRST) detection, and gradient-based attribution for deep networks.

2.1 Single-frame infrared small target detection

Traditional SIRST detection methods mainly include spatial-filtering-based methods [34], [35], [36], [37], sparsity-and-low-rank-based methods [38], [39], [40], [41], and human-visual-system-based (HVS) methods [42], [43], [44], [45]. Recently, deep learning-based methods achieved better detection performance than traditional ones due to their strong feature learning abilities. Drawing inspiration from traditional methods, some networks are integrated with domain-specific priors for robust SIRST detection, such as the local spatial saliency differences between targets and backgrounds [46], [47], [48], the sparsity of targets and the low rankness of backgrounds [32], [49], [50]. Dai et al. [46] incorporated local contrast measure [51] into the network. Zhang et al. [47] converted the local contrast measure into a nonlocal quadrature difference measure in deep feature space. Zhou et al. [49] developed a deep unrolling network to exploit the low-rankness of backgrounds and the sparsity of targets. Liu et al. [50] combined low rankness and local smoothness prior of IRSTs to better model the background. Wu et al. [32] formulated the IRST detection task as sparse target extraction, low-rank background estimation, and image reconstruction in a relaxed robust principle component analysis model. For performance improvements, some methods combine IRST properties with the attention mechanism [52] to learn correlations among pixels and pay attention to key features. Li et al. [30] cascaded channel and spatial attention module to enhance multi-level features. Zhang et al. [53] proposed a Taylor finite difference-inspired edge block and a two-orientation attention aggregation block to highlight the shape of the targets. Zhang et al. [54] introduced a context mixer decoder based on spatial and frequency attention to modulate the low-level features for feature representation enhancement. Liu et al. [18] adopted the self-attention mechanism of Transformer to learn the non-local correlation of image features. Beside feature extraction, Liu et al. [50] proposed a scale and location sensitive loss function to boost detection performance. Zhang et al. [55] integrated textual information with image modalities to enhance IRSTD performance. Zhang et al. [56] proposed a wavelet-based pruning method for efficient detection. Li et al. [57] enabled the convolution to explicitly and differentially model high-frequency components. Moreover, some researchers investigated the weakly-supervised detection [58], [59], [60], in which the pseudo-masks are generated based on the local saliency of the IRST. SIRST detection is applied to locate the salient targets, but is not suitable for targets with low SNR.

2.2 Multi-frame infrared small target detection

MIRST detection aims at exploiting both spatial and temporal information from input image tensors, and is expected to achieve better and more robust performance than SIRST

detection. Existing MIRST detection methods can be categorized into filtering-based methods [61], [62], [63], [64], [65], [66], [67], sparsity-and-low-rank-based methods [7], [29], [68], [69], [70], [71], [72], and deep learning-based methods [6], [23], [24], [73], [74], [75], [76].

Filtering-based methods are the earlier developed methods, which mainly uses the temporal information among frames. Reed et al. [61], [62] proposed a series of 3D matched filtering methods to highlight the targets through spatial filtering and temporal energy accumulation. Gao et al. [63] proposed a time variance and patch contrast filter to improve the local contrast of targets. Li et al. [64] achieved MIRST detection by using the directional morphological filtering and spatial-temporal semantic information (e.g., the consistency of targets and the fluctuation of clutters). Mooney et al. [77] and Silverman et al. [78] early recognized the effectiveness of temporal profile filtering for detecting weak point IRSTs, making the “temporal profile” concept popular in IRST detection. Building on this foundation, some temporal profile-based filters [65], [66], [67] have been developed. These filters offer advantages in both detection performance and computational cost compared to other filtering-based methods.

Sparsity-and-low-rank-based methods decompose the low-rank input tensors, which is developed from the corresponding methods in SIRST detection, by further considering the time dimension. Sun et al. [5] extended the spatial matrix to a spatial-temporal tensor to fit highly heterogeneous scenes. Wu et al. [29] introduced a 4-D infrared image patch tensor structure to accommodate both local and global features in the space and time dimension. Liu et al. [72] proposed graph Laplacian regularization to describe the low rankness of backgrounds, which improved the speed of the detection and the robustness of the method.

Deep learning-based methods mainly include spatial-temporal saliency feature-based methods [22], [23], [24] and motion feature-based methods [6], [73], [74], [79]. Spatial-temporal saliency feature-based methods mainly utilize the saliency information of targets in a single frame and the difference information between two frames. Du et al. [22] built an end-to-end spatial-temporal feature extraction and target detection framework based on an energy accumulation enhancement mechanism between frames. Yan et al. [23] proposed to use a temporal multiscale feature extractor to obtain spatial-temporal features from multiple time scales. Tong et al. [24] established the spatial-temporal dependencies among frames by using temporal information. Motion feature-based methods improve the IRST detection performance especially in suppressing false alarms by using the continuity of the target motion in the space-time dimension. Sun et al. [73] modeled the local smoothness and global continuous characteristic of the target trajectory as short-strict and long-loose constraints, to detect the true targets from numerous target candidates. Guo et al. [79] applied a trained classification convolution neural network to identify the trajectory mask images, and linked adjacent trajectory segments physically. The above methods leverage the motion features only in track association, but not in target candidates detection, which increases the difficulty of trajectory management. Then, Li et al. [6] designed a direction-coded convolution block to encode the motion direction into features and extract the motion information of targets for MIRST detection. Zhang et al. [74] proposed a hybrid modeling method with smoothed-particle hydrodynamics for motion trajectory simulation and Markov decision processes

for optimal action selection. In conclusion, the above methods are based on the spatial and temporal saliency information and motion information. However, these information are weak and broken under extremely complex conditions (e.g., dim targets and strong interference), resulting in failed detection. Moreover, there are amounts of computational redundancy to extract this information due to the large receptive field of the network and the sparsity of the target.

2.3 Gradient-based attribution for deep networks

Attribution [80], [81] is an approach of network interpretation to understand the predictions given by the models, through highlighting input features (pixels in the case of image input) that strongly influence the network outputs. Gradient-based attribution [82], [83], [84], [85] is a typical branch of attribution, which exploits the information flow pathway in deep networks and generate saliency maps [86]. Here, we review two major types of approaches, i.e., vanilla gradient-based methods [82], [83], [84], [87], [88], [89] and integrated gradient-based methods [26], [27], [90], [91].

Vanilla gradient-based methods explain the models by directly using gradients. Gevrey et al. [82] and Simonyan et al. [83] proposed to consider both the gradients and their variants to explain the models. Zeiler et al. [84] proposed the deconvolutional network (DeconvNet) to approximately visualize the input stimuli that excite individual feature maps in a specified layer. Considering that DeconvNet performs poorly without max-pooling layers, Springenberg et al. [87] proposed the guided backpropagation algorithm including an additional guidance signal from the higher layers. Zhou et al. [88] developed class activation mapping (CAM) to identify discriminative regions for predictions via the linearly weighted summation of activation maps from the last convolution layer. Then, Selvaraju et al. [89] introduced gradient-weighted CAM (Grad-CAM) to use the gradients w.r.t. the activation map to measure the channel importance. Based on Grad-CAM, a series of methods were proposed, e.g., Grad-CAM++ [92], Smooth Grad-CAM++ [93]. However, this type of methods may fail in some cases due to the saturation issue of the gradients [26], [90].

Integrated gradient-based methods explain the models by accumulating gradients along a specified path from a baseline [85], [94], [95] to the input pixels. Sundararajan et al. [26] first proposed axiomatic attribution (i.e., integrated gradients) to address the problem of gradient saturation. Xu et al. [27] proposed the blur integrated gradients to explain the prediction by accumulating gradients from both the frequency and space domains. Sturmfels et al. [90] proposed alternative choices for baselines in integrated gradients, considering that the all-zero baseline is unsuitable for black targets. Miglani et al. [91] observed that the accumulated gradients within the saturation regions also have a non-negligible contribution on saliency maps, but not on improving the model prediction.

However, all the above methods are aimed at classification tasks, and only a few works consider explaining dense-prediction networks [96], [97], [98]. Hoyer et al. [96] proposed the first attribution method to generate grid saliency maps for dense prediction networks. Gu et al. [97] succeeded in applying the integrated gradients to super-resolution networks by building suitable baseline images and a gradient detector. Chrabaszcz et al. [98] introduced AGG²EXP to aggregate fine-grained voxel attributions of 3D segmentation models. However, the attribution of IRST detection networks lacks exploration.

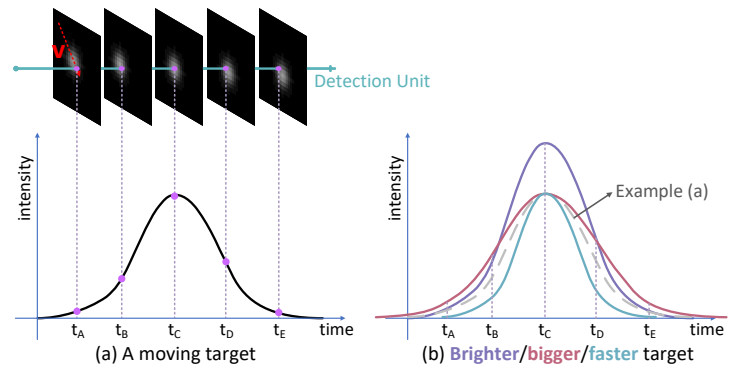


Fig. 3. Toy examples of the temporal profiles of different targets.

3 METHODOLOGY

In this section, we present the more essential information for IRST detection especially under complex conditions, and introduce our deep temporal probe network in detail. We first analyze the temporal profile information supporting the detection, and then verify the importance of the information through building the first attribution tool and applying it to IRST detection networks. Finally, we describe our temporal probe mechanism and our DeepPro with only temporal calculations.

3.1 Temporal profile information

Given the loss of spatial saliency, local temporal saliency, and motion characteristics of some IRSTs, we explore essential statistical information at the signal level to support target predictions. In this part, we shift focus from the spatial domain and the short-term spatial-temporal domain to the temporal profile. Then, we delve into what information is in the temporal profile and how it supports the detection of IRST.

3.1.1 Global temporal saliency

We analyzed the waveform characteristics of target signals in the temporal profile. Given that the velocity of small targets exhibits minimal variation over the short time they pass through a detection unit completely, we modeled the moving targets in uniform linear motions. In Fig. 3, we display some toy examples of the temporal profiles of the targets with different intensities, sizes, and speeds. The fluctuation of a target signal G is regular with a single up-down process. As a target moves into and out of a detection unit, the target intensity in this unit first increases and then decreases. The amplitude of G is related to the maximum intensity of the part of the target passing through the detection unit, and a brighter target usually has a higher peak value. Besides, the width of G is related to the size and speed of the target. Bigger targets typically exhibit wider signals in the temporal profile, while targets with higher speeds under the same conditions produce narrower signals. Therefore, assuming that the fluctuations of other signals can be ignored, the target signals are saliency in the temporal profile.

To investigate whether the saliency remains in the practical complex scenarios, we visualized the temporal profiles of real target signals with different SNR values, and compared them with the spatial domains of the same targets. The visualizations are shown in Fig. 4. It can be seen that the fluctuations of the target signals are consistent with those of the toy examples, i.e., first increasing and then decreasing. When the target SNR is higher than 3, the target is significant in both the spatial domain and the temporal profile. However, when the target SNR is

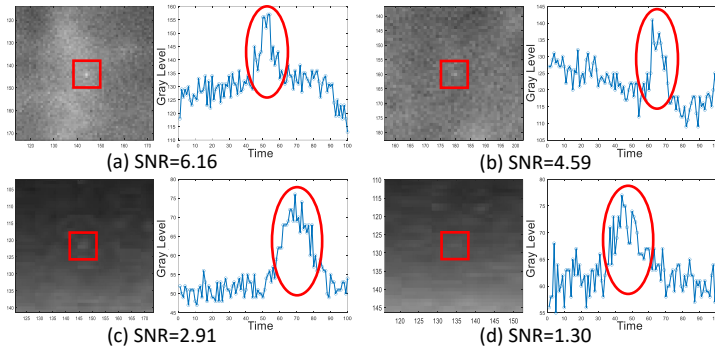


Fig. 4. Spatial and temporal-profile visualizations of targets in real scenes with noise and clutter. The red annotations mark the targets.

less than 3, the target is so dim that it is barely visible in the spatial domain, but is still prominent in the temporal profile. This demonstrates that the temporal profile holds the global temporal saliency of the target signals, and is more robust to distinguish the targets from strong noise and clutter than other domains.

3.1.2 Correlation information

As the intensities of noise and clutter signals increase, the global temporal saliency becomes weak and damaged (e.g., Fig. 4(d)). Therefore, we deeply extract other essential distinctive characteristics between noise N , clutter B , and target G in the temporal profile.

Firstly, to further analyze the differences between target and noise signals, we mixed target and noise, and analyzed the correlation function of the mixed signals (i.e., $f_{G+N}(t)$) which quantifies the similarity of signals at different time points. We added the same target signal to the simulated noise signals² with different standard deviations σ_N , as shown in Fig. 5(a1). The target signal still has significance and the same up-down process when the noise is relatively weak (e.g., $\sigma_N = \frac{1}{4}$) in the time dimension. As the noise becomes stronger (e.g., $\sigma_N = \frac{1}{2}$), these features gradually become less pronounced. Then, we analyzed the correlation of $f_{G+N}(t)$ to find if there are still some distinctive characteristics. The correlation function $R_{G,N}(\tau)$ is calculated by measuring the inner product of $f_{G+N}(t)$ and $f_{G+N}(-t)$ under the time delay τ , i.e.,

$$R_{G,N}(\tau) = \int_{-\infty}^{+\infty} f_{G+N}(t) \cdot f_{G+N}(-t + \tau) dt = f_{G+N}(\tau) * f_{G+N}(\tau). \quad (1)$$

After smoothing, the correlation function is shown in Fig. 5(a2). When there is no time delay (i.e., $\tau = 0$), the correlation value is maximum since the target is autocorrelated in the time dimension. When $\tau \neq 0$, the correlation value is close to zero. This demonstrates that the target signal is uncorrelated with the noise signal, which is also supported in [100]. Even for strong noise (e.g., $\sigma_N = \frac{1}{2}$), the target still maintains high autocorrelation characteristics and distinctiveness from the noise in the temporal profile. Therefore, *the temporal profile holds the correlation information between target and noise, which is advantageous to predict the target signal even mixed with extremely strong noise.*

Then, to investigate the differences between target and clutter, we mixed target and different clutters (e.g., land clutter and moving cloud clutter), and analyzed the correlation function

of the mixed signals (i.e., $f_{G+B}(t)$). At first, we got different clutters from the real video scenarios. After decentralizing and scaling, we added the fixed-intensity target signal to a series of clutter signals with different standard deviations σ_B ³, as shown in Fig. 5(b1) and (c1). The typical clutters usually include low-frequency signals (e.g., slowly-changing cloud in c1) and high-frequency signals (e.g., camera blind elements and flickering background in b1). High-frequency signals are more likely to be falsely detected as targets due to their spatial and temporal saliency. When the clutter signal in the time dimension is not strong, there are some obviously distinctive characteristics between targets and clutters. Usually, the fluctuations of the clutter are dense, and they are correlated with each other on the amplitude in a period but uncorrelated on the occurrence moment. However, these characteristics are still inadequate for the detection when the clutter has a high standard deviation. Following the above analysis, we calculated the correlation function $R_{G,B}(\tau)$ of $f_{G+B}(t)$, i.e.,

$$R_{G,B}(\tau) = \int_{-\infty}^{+\infty} f_{G+B}(t) \cdot f_{G+B}(-t + \tau) dt = f_{G+B}(\tau) * f_{G+B}(\tau). \quad (2)$$

The smoothed correlation function is shown in Fig. 5(b2) and (c2). It can be observed that the target retains significant autocorrelation characteristics even if mixed with high-intensity land clutters or cloud clutters. Besides, the target signal is usually uncorrelated with the clutter signal as the correlation value at $\tau \neq 0$ is close to zero. Therefore, *the temporal profile contains a lot of correlation information between target and clutter, which is reliable to detect the target from even high-intensity clutters.*

According to the above analyses, target, noise, and clutter signals exhibit different characteristics in the temporal profile. Specifically, target signals are autocorrelated with a regular process and are irrelevant to noise and clutter signals. These distinct characteristics make the temporal profile more robust than other domains for IRST detection.

3.2 Attribution verification

From the theoretical analysis, we conclude that the temporal profile information is very important for IRST detection. As for the deep networks learning features from numerous labeled data, it is still unclear whether the temporal profile information is really essential for model predictions, whether it is more robust than the information in the spatial domain and the short-term spatial-temporal domain, and whether every moment in the temporal profile is important. In this section, we verify the importance of the temporal profile information in IRST detection networks.

3.2.1 Attribution approach for IRST detection networks

Before introducing the attribution approach in this work, we briefly summarize the preliminary [26] in classification networks. Let a function $F : \mathbb{R}^n \rightarrow [0, 1]$ and an input $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$ denote a detection model and its input. Assume that a_i is the contribution of x_i to the prediction $F(\mathbf{x})$. Then, the attribution of the prediction at input $\mathbf{x}$ is defined as

$$A_F(\mathbf{x}, \mathbf{x}') = (a_1, \dots, a_n) \in \mathbb{R}^n. \quad (3)$$

The prediction attribution is relative to a baseline input $\mathbf{x}'$ which simulates the absence of important features and is used

2. According to [99], [100], [101], the combination of various noises in real scenes follows a Gaussian distribution, and can be modeled by Gaussian white noise.

3. Note that the standard deviation of clutter in the time dimension is different from its fluctuation intensity in the space dimension. Usually, the temporal standard deviation of a clutter is smaller than its local spatial fluctuation since the background is slowly changing.

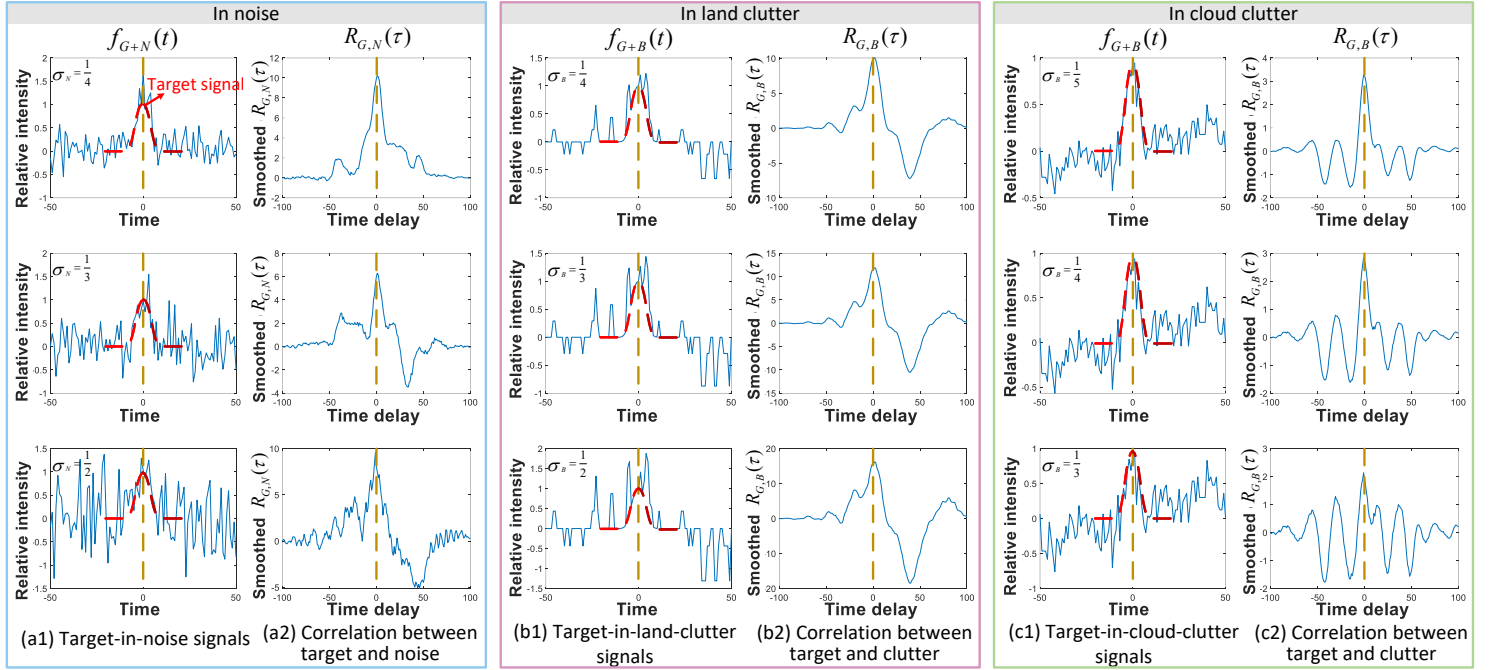


Fig. 5. Visualizations of the temporal profiles of target-in-noise(-clutter) signals and correlation analyses between noise, clutter and target in the temporal profile. The target signals are same with fixed intensity (maximum is 1), and added to noise and clutter signals with different standard deviations (i.e., σ_N and σ_B , respectively).

to provide a comparative reference. Consequently, the model prediction on a baseline input is usually zero (i.e., $F(\mathbf{x}') = 0$). The necessity of a baseline has been proved in [94], [95]. Generally, the baseline input is an all-zero matrix or a matrix without information that can affect the prediction. Integrated gradients (IG) [26] is a typical attribution approach, and is formulated as

$$IG(\mathbf{x}) = (\mathbf{x} - \mathbf{x}') \int_0^1 \frac{\partial F(\mathbf{x}' + \alpha \times (\mathbf{x} - \mathbf{x}'))}{\partial \mathbf{x}} d\alpha. \quad (4)$$

Here, $\frac{\partial F(\mathbf{x})}{\partial \mathbf{x}}$ is the gradient of $F(\mathbf{x})$ w.r.t. the input, which measures the local impact of input variations on the output within a narrow vicinity around $\mathbf{x}$.

Based on integrated gradients, we developed the first prediction attribution analysis tool for IRST detection networks. Unlike classification networks, segmentation-based IRST detection networks generate the prediction of each spatial pixel in the input $\mathbf{S} \in \mathbb{R}^{T \times H \times W}$, and can be represented as $D : \mathbb{R}^{T \times H \times W} \rightarrow \mathbb{R}^{H \times W}$. T denotes the temporal length of the input, while $H \times W$ denotes the input's spatial size. To interpret D , we attribute the prediction of a single target globally. Specifically, we need a converter to quantify the prediction of a target, $C : \mathbb{R}^{H \times W} \rightarrow \mathbb{R}$. In this work, we achieved the converter by summing the predictive scores of all pixels in this target. That is,

$$C(D(\mathbf{S})) = \sum_{i=1}^H \sum_{j=1}^W D(\mathbf{S}) \odot M_k, \quad (5)$$

where M_k represents the mask of the k^{th} target, and $\odot$ donates Hadamard product.

To generate the attribution map of $C(D(\mathbf{S}))$, we accumulated the gradients along a specified path from a baseline input $\mathbf{S}'$ to $\mathbf{S}$. The path function is represented as $\gamma(\alpha) : [0, 1] \rightarrow \mathbb{R}^{T \times H \times W}$. Then, the attribution for IRST detection models is formulated as

$$A_{D,C}^{\gamma}(\mathbf{S}, \mathbf{S}') := \int_0^1 \frac{\partial C(D(\gamma(\alpha)))}{\partial \gamma(\alpha)} \times \frac{\partial \gamma(\alpha)}{\partial \alpha} d\alpha, \quad (6)$$

where $\mathbf{S}' \in \mathbb{R}^{T \times H \times W}$. Since the targets always present as high-frequency components in the input, the baseline should avoid containing high-frequency components. Meantime, arbitrary fluctuation in the background can affect the target prediction. Therefore, the constant tensor is suitable as a baseline for IRST detection, and the constant should be lower than the gray average of the target since integrated gradients fail to highlight black targets [81]. The path used in this work is a straight line, i.e.,

$$\gamma(\alpha) = \mathbf{S}' + \alpha(\mathbf{S} - \mathbf{S}'). \quad (7)$$

With this attribution approach, the predictions of arbitrary segmentation-based IRST networks can be attributed.

3.2.2 Attribution analysis

We analyzed the attribution results of a typical CNN-based MIRST detection network. We selected a widely used open source network, i.e., STDMA Net [23], which takes 19 adjacent reference frames and a predicted frame as its input and detects the targets in the predicted frame.

We set up different seeds (e.g., 1202 [23], 3407 [102], 42, 500, 1500, 2000, 2500, 3000) to train the network and observe the attribution results in different scenes. We first visualized the distribution of the input pixels in reference frames that significantly influence the prediction. The visualization is shown in Fig. 6, from which we observe Phenomenon 1 and verify that the temporal profile information is more important than the spatial information.

Phenomenon 1. *The influential pixels in reference frames are mainly distributed around the temporal profile of the target. The core influence region is close to a cylindrical zone along the time axis.*

As aforementioned, many IRST detection methods expand the spatial receptive field and supplementing the limited temporal continuous receptive field for more information. Following these thoughts, the influential pixels should be distributed over a wide spatial-temporal (especially spatial)

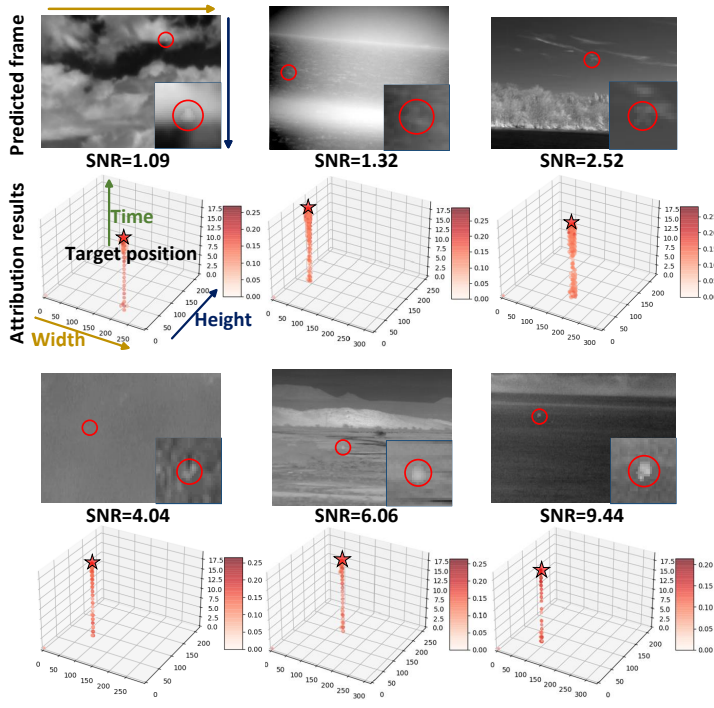


Fig. 6. Attribution visualizations of reference frames at targets with different SNRs. For the predictions of the salient targets and the dim targets, the influential pixels are almost concentrated on the temporal profile of the target.

field covering the target trajectory. However, Fig. 6 shows that the influential pixels in reference frames are concentrated on the temporal profile of the target in the predicted frame. The influence region is small in the space dimension but covers all reference moments, and the region does not cover the target trajectory since the region is along the time axis. The results reveal that the thoughts in existing methods are one-sided under no matter simple scenes or extremely complex conditions, and *the temporal profile information of the target is more important for its prediction.*

Since the phenomenon demonstrates the significance of the temporal profile, a natural question arises: Is the influence of each reference frame on the predicted frame related to its temporal distance from the predicted moment? Therefore, we statistically assess the average influence (across a whole sequence) of different reference frames on the prediction, and the results are shown in Fig. 7. From the results, we observe Phenomenon 2 and verify that long-term temporal information is important.

Phenomenon 2. *The importance of a reference frame changes over time, following a U-shaped curve. Distant information is crucial for predictions like recent information in a period.*

Researchers conventionally feed adjacent frames into networks to augment temporal context. There is a potential cognition that temporally proximate information is more important, and temporally distant information is not important. In this way, the curve should increase monotonically. However, there is something different from the above intuition in practical curves. That is, the importance degrees of 19 reference frames first decrease and then increase over time, exhibiting a U-shaped pattern. The difference reveals that a few adjacent frames are not enough, and distant information is also important for model prediction. In other words, *the long-term variations of the signals in the temporal profile are crucial forIRST detection.*

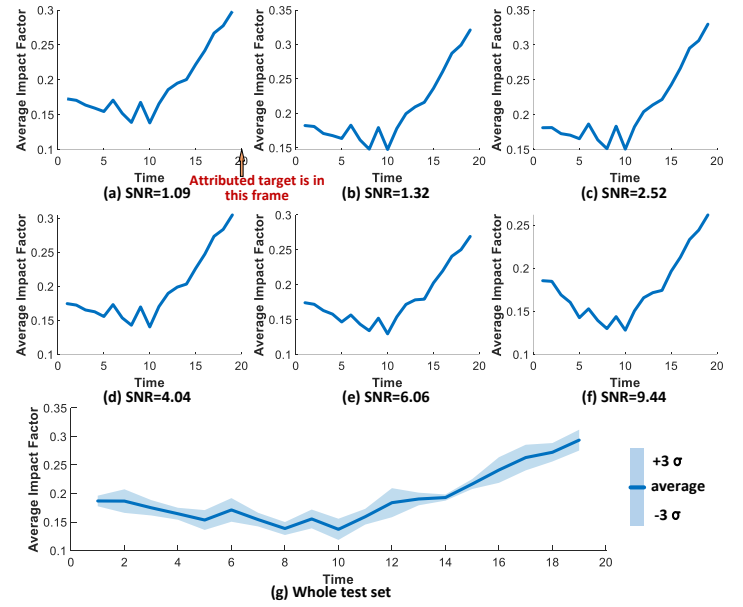


Fig. 7. Temporal variations of the average influence and the influence range of all random seeds in different scenes. (a)-(f) show the temporal variations on the targets with SNR of 1.09, 1.32, 2.52, 4.04, 6.06, and 9.44, respectively. (g) illustrates the temporal variation in the whole test set.

3.3 Temporal probe mechanism

Inspired by prior research, we remodel theIRST detection task as a one-dimensional (i.e., time-dimension) signal anomaly detection task, treating target signals as temporal anomalies. As previously analyzed, it is important for the networks to extract long-term temporal correlation information for this task. Ordinary 3D convolution operations focus only on short-term features and require stacking multiple convolution layers to achieve long temporal coverage. However, the temporal receptive field gradually expands as the number of stacked layers increases, and thus a amount of feature extraction computation is still based on short-term features. Moreover, multiple 3D convolutional layers bring substantial computational overhead for large-scale feature tensors. Therefore, in this part, we propose a temporal probe mechanism (TPro) to efficiently extract long-term temporal correlation features and to probe whether the temporal features of a pixel exist anomaly. The structure of TPro is shown in Fig. 8.

As theIRST detection task is remodeled as a one-dimensional (i.e., time-dimension) signal anomaly detection task, TPro is designed to focus on the long-term temporal features of an arbitrary pixel. The input is some basic features $F^b \in \mathbb{R}^{C \times T \times H \times W}$ of a series of input frames (with the temporal length of T and the spatial size of $H \times W$), which are also the outputs of the models before TPro in a network. Its output $F^{tp} \in \mathbb{R}^{C \times T \times H \times W}$ is the features containing some temporal profile information. According to the analysis in Section 3.1, the target, background, and noise signals have different characteristics in the temporal correlation. Therefore, we used the learnable statistics-based correlation matrix (SCorM) $W^{T \times T} \in \mathbb{R}^{T \times T}$ to perceive temporal profile information and distinguish target signals from background signals. A matrix can learn a set of statistical properties that describe the correlation between signals at different moments. To fully learn these properties, it is essential to apply multiple SCorMs (i.e., $W_1^{T \times T}, W_2^{T \times T}, \dots, W_m^{T \times T}, m > 1$). The operation details of TPro are described below.

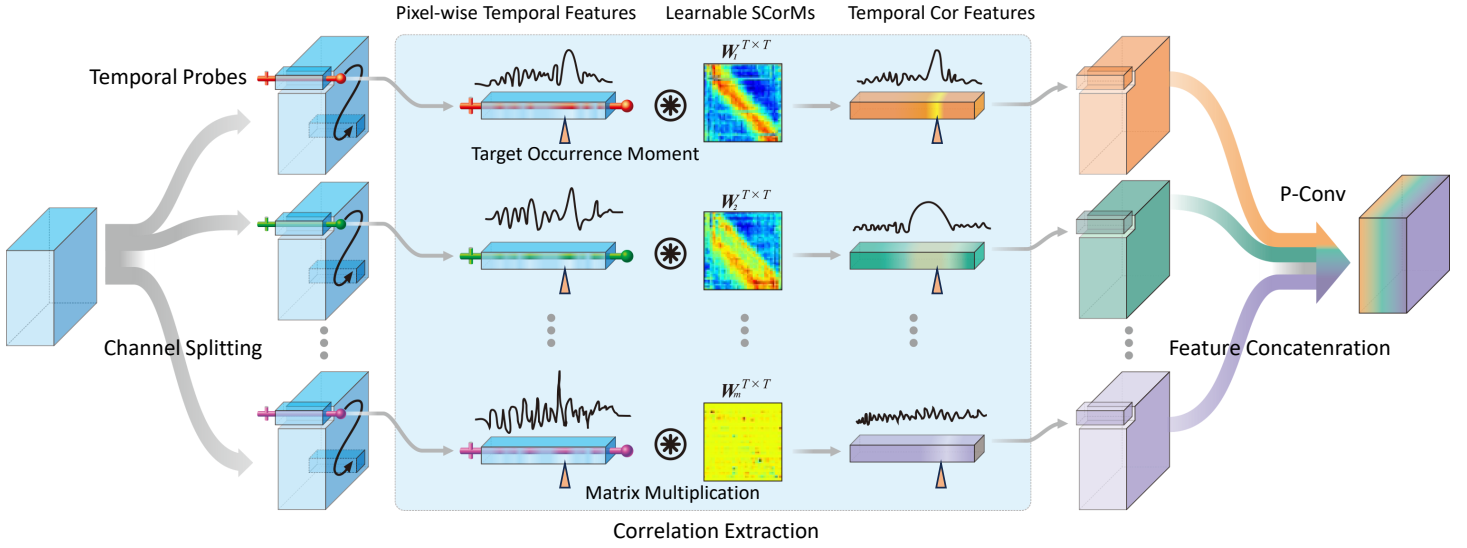


Fig. 8. Visualization of the structure of TPro. Firstly, temporal probes are employed to pull out complete temporal features of a single pixel from the split input features. Then, multiple learnable SCorMs are applied to the gotten features to extract temporal correlation features. This process is performed pixel-wise across all split input features. Finally, all temporal correlation features are concatenated and fused to generate output features. Due to the learned correlationship among signals, target features are enhanced, and background features are suppressed. Notably, the entire pipeline strictly excludes any space-dimension computations.

As shown in Fig. 8, the input basic features are first divided into m features along the channel dimension, i.e., $F_i^b \in \mathbb{R}^{\frac{C}{m} \times T \times H \times W}$ ($i = 1, 2, \dots, m$). To perform computations on the long-term temporal features of a pixel, temporal probes are utilized to individually extract the complete temporal features of each pixel, i.e., $F_i^b(x, y) \in \mathbb{R}^{\frac{C}{m} \times T \times 1 \times 1}$ ($x \in [1, W], y \in [1, H]$). Subsequently, the correlation matrix is applied to the extracted features for the temporal correlation features $F_i^{cor}(x, y) \in \mathbb{R}^{\frac{C}{m} \times T \times 1 \times 1}$. The process can be described as

$$F_i^{cor}(x, y) = F_i^b(x, y) \cdot W_i^{T \times T}, \quad (8)$$

where $W_i^{T \times T}$ represents the i^{th} correlation matrix. After repeating the process, the temporal correlation features (i.e., F_i^{cor}) of all pixels are obtained. To fuse temporal correlation information from different channels and correlation matrixes, a pointwise convolution [103] (P-Conv, i.e., $1 \times 1 \times 1$ convolution) layer $\phi(\cdot)$ is applied to these features (i.e., $F_1^{cor}, F_2^{cor}, \dots, F_m^{cor}$) after concatenation along the channel dimension. This process can be described as

$$F^{tp} = \phi([F_{i,1}^{cor}, F_{i,2}^{cor}, \dots, F_{i,m}^{cor}]), \quad (9)$$

where $\phi(\cdot)$ includes convolution operation, batch normalization, and ReLU activation [104]. Finally, the long-term temporal correlation features are generated with highlighting the anomaly signals.

3.4 Deep temporal probe network for IRST detection

The superior temporal profile information and the attribution verification inspire us to develop a new framework that emphasizes the temporal profile information. Therefore, based on TPro, we propose the deep temporal probe network called DeepPro. Our DeepPro consists of temporal feature extraction module (including TPro) and fusion & segmentation module. There are three similar level structures operating in parallel, as shown in Fig. 9. The input is a long sequence $S \in \mathbb{R}^{T \times H \times W}$, and is first sent to the temporal feature extraction module for temporal features. After feature fusion and segmentation

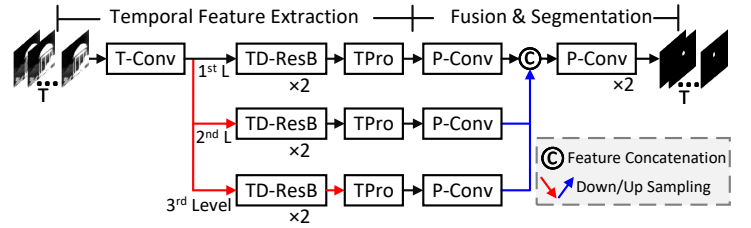


Fig. 9. An overview of our DeepPro. There are three similar levels in our DeepPro. Note that this is the first IRST detection network with calculations (e.g., multiplications and additions) only in the time dimension.

operations, i.e., a series of $1 \times 1 \times 1$ convolution layers, the final confidence maps of all input frames are generated, i.e., $C \in \mathbb{R}^{T \times H \times W}$. There is no deep spatial-temporal feature extraction structure employed in most MIRST detection networks for a large spatial and short temporal receptive field, but only few temporal convolutions and TPro for long-term temporal information (i.e., temporal profile information). Our DeepPro achieves unprecedented simplicity in IRST detection, which is the first IRST detection network with calculations only in the time dimension. The details of the temporal feature extraction are described below.

Specifically, tens of frames S are first fed to a temporal convolution (T-Conv) layer⁴ with a kernel $W_T \in \mathbb{R}^{l \times 1 \times 1}$ ($l = 5$). Then, the features are sent to a few identical temporal difference residual blocks (TD-ResBlock) to roughly extract the basic temporal features F^b . Each TD-ResBlock contains a temporal difference convolution (TD-Conv) layer $\psi(\cdot)$ with a $3 \times 1 \times 1$ kernel W_{TD} and a temporal dilation of 2, a T-Conv layer ($l = 3$), a P-Conv layer and the skip connection. TD-Conv serves to maintain and emphasize the temporal saliency of targets roughly. Inspired by the temporal-difference design in [105], it accomplishes this by applying convolution to multiple temporal difference features (i.e., $[f_t - f_{t-1}, f_t - f_{t+1}]$), which

4. Each convolution layer consists of a convolution operation, batch normalization, and ReLU activation.

can be described as

$$\begin{aligned}\psi(\cdot, \mathbf{W}_{\text{TD}}) &= \sum_{t=1}^l w_t \cdot (2 \cdot f_t - f_{t+1} - f_{t-1}) \\ &= \sum_{t=1}^l (2 \cdot w_t - w_{t+1} - w_{t-1}) \cdot f_t,\end{aligned}\quad (10)$$

where f_t is the feature value of the t^{th} ($t = 1, 2, 3$) dimension in the time dimension within a sliding window. w_t is the t^{th} weight of the convolution kernel $\mathbf{W}_{\text{TD}}$.

To obtain the basic temporal features of the entire target region, the above process is performed multiple times at different levels. The only difference between these levels lies in the size of the features they process. The sizes are different due to the different number of max-pooling operations applied in a specific level. After multiple above processes, a series of basic temporal features are obtained, i.e., $F_j^b \in \mathbb{R}^{C \times T \times H \times W}$ ($j = 1, 2, 3$). They are sent to TPro respectively for long-term temporal correlation feature extraction. Finally, various features containing temporal profile information are acquired, i.e., $F_1^{tp}, F_2^{tp}, F_3^{tp}$.

4 EXPERIMENTS

4.1 Experimental setup

4.1.1 Datasets

We evaluated our method on the NUDT-MIRSDT dataset [6], the IRDST dataset, [106] and the real scenarios from the RGBT-Tiny dataset [107]. The IRDST-simulation dataset contains 102077 frames from 316 sequences. The NUDT-MIRSDT dataset contains 120 sequences, and each includes 100 frames. The test set of the NUDT-MIRSDT dataset is divided into two subsets according to their SNR. One consists of 8 sequences with SNR lower than 3, and the other consists of 12 sequences with SNR ranging from 3 to 10.

To further verify the robustness of our DeepPro to dim targets and complex scenarios, we built a dataset with strong noise by adding high-intensity noise to the NUDT-MIRSDT dataset, called NUDT-MIRSDT-HiNo dataset. The noise model is described in the Appendices. Detailed information on the NUDT-MIRSDT-HiNo dataset is reported in Table 1. Fig. 10 shows the effect of the strong noise on the targets. The noise makes the bright targets dim and makes the dim targets invisible in the space dimension.

TABLE 1

Parameters related to noise of the datasets. The standard deviation of noise (σ_n), the distribution range of the multiplicative noise (σ_g), the standard deviation of the additive noise (σ_o), the pattern of non-uniformity, and the average SNR values of the whole dataset and the test set.

Datasets	σ_n	σ_g	σ_o	non-uniformity	SNR	SNR (test)
NUDT-MIRSDT	0.1	0	0	none	11.67	7.00
NUDT-MIRSDT-HiNo	8.0	0.15	1.3	pixel-wise	3.86	3.24

4.1.2 Metrics

We followed [6] to use the probability of detection (P_d), false alarm rate (F_a), and area under curve (AUC) as quantitative metrics for the evaluation of IRST detection performance. P_d measures the ability of a method to detect targets. F_a measures the ability of a method to correctly identify the background, such as clutter. AUC is the area value under the receiver operating characteristics (ROC) curve [108], [109], reflecting the detection performance under different thresholds. Besides,

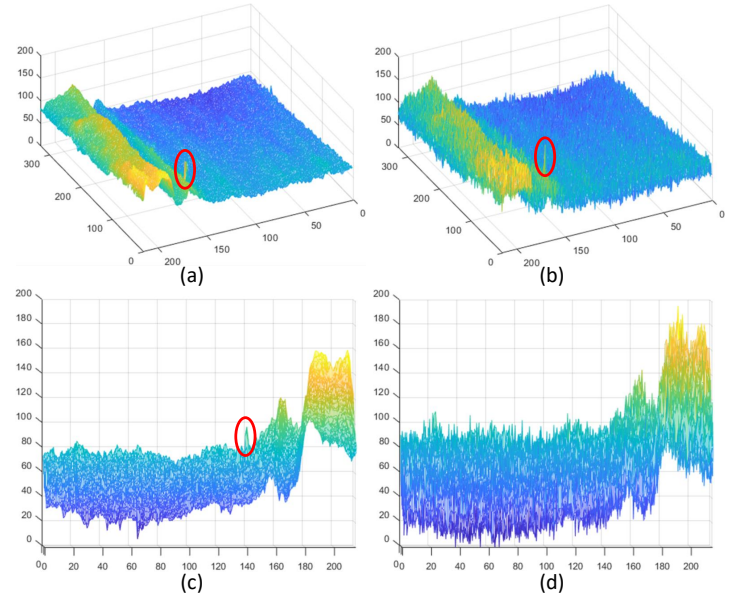


Fig. 10. 3D visualization of same targets in the NUDT-MIRSDT dataset and the NUDT-MIRSDT-HiNo dataset. (a) The bright target in the NUDT-MIRSDT dataset. (b) The same bright target in the NUDT-MIRSDT-HiNo dataset. (c) The dim target in the NUDT-MIRSDT dataset. (d) The same dim target in the NUDT-MIRSDT-HiNo dataset.

the number of parameters (#Params), GFLOPs, and FPS are measured to evaluate the scale, computational complexity, and computational efficiency of the model, respectively. Specifically, GFLOPs and FPS are calculated based on input images of size 256×256 . In the following result tables, P_d is expressed in units of 10^{-2} , F_a in 10^{-5} , #Params in millions (M), GFLOPs in GFLOPs per frame, and FPS in frames per second.

4.1.3 Implementation details

All deep learning-based methods were implemented in PyTorch on a computer with an Intel Xeon Gold 6328H CPU @ 2.80GHz and two Tesla V100s PCIe 32GB GPUs. In detail, the networks were trained for 32 epochs with a batch size of 4. We used the Adam optimizer [110] with an initial learning rate of 0.001. The learning rate decayed after every 10 epochs with a decaying rate of 0.7. We initialized the weights of the convolution layers using the Kaiming method [111].

The input of our DeepPro was set to 40 consecutive frames ($T = 40$) in a sequence, and the number of temporal probes was set to 4 ($m = 4$). For data augmentation, we randomly selected one frame as the first frame in the input. If the number of subsequent frames is less than T , images with all zero values are filled. We adopted random cropping with a patch size of 128×128 , and ensured that the probability of the existence of targets in a sample is higher than 75%. The Soft-IoU [112] loss function was used for loss calculation. In the inference phase, we set the repetition rate at 10%⁵ to ensure that the targets at both ends of the input sequence can be well detected. When the remaining frames are less than T , overlapping windows are used to form complete frames. The final detection results for frames input multiple times through overlapping regions are combined using a union operation.

5. For example, the first sample is continuous frames from the 1^{st} to the T^{th} , and the next sample is continuous frames from the $((1 - 10\%)T + 1)^{\text{th}}$ to the $((1 - 10\%)T + T)^{\text{th}}$.

TABLE 2

Detection results achieved by different state-of-the-art methods on the NUDT-MIRSDT dataset and the NUDT-MIRSDT-HiNo dataset. The best results are in bold, and the second-best results are underlined. *SF* and *MF* refer to single-frame and multi-frame methods, respectively.

Methods		$SNR \leq 3$		NUDT-MIRSDT			NUDT-MIRSDT-HiNo			#Params (M)	FPS	
		P_d	F_a	P_d	F_a	AUC	P_d	F_a	AUC			
Traditional Methods	MF	MSLSTIPT [5] (TGRS'20)	4.16	21.70	18.97	15.37	0.9404	3.93	73.76	0.9185	-	0.17
		IMNN-LWEC [70] (TGRS'22)	0.00	7.22	26.43	10.74	0.6734	4.97	83.28	0.5394	-	30.31
		SRSTT [7] (TGRS'23)	69.94	6.12	90.63	3.35	0.9989	4.34	55.04	0.5358	-	0.06
		4DST-BTMD [28] (TGRS'23)	41.58	23.45	44.77	74.95	0.8488	4.80	77.29	0.6651	-	26.09
		STRL-LBCM [71] (TAES'23)	5.48	85.53	19.03	34.05	0.5972	2.55	77.78	0.5238	-	0.87
		4D-TR [29] (TGRS'23)	55.77	2.55	55.70	3.19	0.9946	4.63	120.16	0.6633	-	0.36
		4D-TT [29] (TGRS'23)	24.95	1.67	30.89	3.21	0.8287	6.94	73.18	0.5347	-	10.82
		NFTDGSTV [101] (TGRS'23)	1.51	32.31	13.77	35.32	0.8613	11.56	43.16	<u>0.9524</u>	-	0.58
Deep-Learning Methods	SF	ACM [113] (WACV'21)	7.75	22.88	51.53	17.52	0.9298	0	-	-	<u>1.592</u>	171.31
		ALCNet [46] (TGRS'21)	3.97	37.10	52.57	25.50	0.8435	36.09	91.99	0.9326	3.457	142.51
		Res-UNet [33] (ITME'18)	15.83	30.32	63.27	40.83	0.9198	35.51	22.55	0.9391	3.656	222.85
		DNA-Net [30] (TIP'22)	23.74	19.23	67.38	15.07	0.8843	49.16	60.98	0.9373	18.787	29.87
		ISNet [53] (CVPR'22)	17.96	8.53	65.99	19.25	0.9123	28.40	90.17	0.9224	4.363	52.25
		UIUNet [31] (TIP'22)	15.12	17.46	61.25	14.42	0.9436	43.67	28.87	0.9246	202.152	12.46
		AGPCNet [114] (TAES'23)	31.76	176.38	55.47	85.56	0.9443	42.452	13655.20	0.7348	49.442	24.13
		MSHNet [115] (CVPR'24)	2.46	78.07	36.78	41.91	0.7966	21.81	44.92	0.6748	16.262	69.35
	MF	SCTransNet [20] (TGRS'24)	23.63	122.24	62.81	74.20	0.9320	20.65	35.37	0.8283	45.304	20.07
		RPCANet [32] (WACV'24)	30.06	81.21	61.13	41.28	0.8694	21.81	198.14	0.8786	2.720	31.47
		ILNet [116] (TAES'25)	17.39	55.89	57.90	34.09	0.7572	34.11	56.53	0.6551	6.322	36.82
		Res-U+DTUM [6] (TNNLS'23)	91.68	2.37	<u>97.46</u>	3.00	0.9967	43.90	4.86	0.9413	1.193	62.65
		STDMA-Net [23] (TGRS'23)	<u>92.82</u>	2.88	96.59	3.40	0.9908	51.65	<u>1.95</u>	0.8766	47.518	18.66
		Res-U+RFR [4] (TGRS'25)	79.39	4.15	93.35	1.95	0.9657	<u>55.64</u>	111.23	0.7839	4.096	102.88
		DQAligner [117] (TGRS'26)	81.29	0.37	94.22	0.15	0.9419	33.41	2.95	0.8022	2.410	10.01
		DeepPro (ours)	95.84	<u>0.52</u>	98.50	<u>0.72</u>	<u>0.9973</u>	59.17	1.76	0.9638	0.197	<u>184.55</u>

TABLE 3

Detection results achieved by different state-of-the-art methods on the IRDST-simulation dataset. The best results are in bold, and the second-best results are underlined.

	Methods	P_d	F_a	AUC	GFLOPs
<i>SF</i>	ALCNet [46]	75.06	1.25	0.9587	0.45
	Res-UNet [33]	82.10	1.34	0.9729	1.30
	DNA-Net [30]	81.44	1.12	0.9724	14.28
	ISNet [53]	82.02	1.06	0.9630	30.63
	UIUNet [31]	87.58	1.69	0.9729	54.50
	AGPCNet [114]	59.48	94.15	0.9529	43.18
	MSHNet [115]	78.01	1.10	1.8413	6.11
	SCTransNet [20]	83.67	6.34	0.9564	10.12
<i>MF</i>	RPCANet [32]	88.06	281.96	0.9686	44.57
	ILNet [116]	41.55	0.33	0.6964	20.97
	Res-U+DTUM [6]	<u>99.48</u>	0	0.9730	6.28
	STDMA Net [23]	98.88	0	<u>0.9978</u>	95.54
	Res-U+RFR [4]	98.90	9.36	0.9950	9.43
	DQAligner [117]	96.86	0	0.9742	27.56
	DeepPro	99.85	<u>0.13</u>	0.9993	<u>1.01</u>

4.2 Comparison to the state-of-the-arts

To demonstrate the effectiveness of our method, we compared our method with state-of-the-art IRST detection methods, including traditional MIRST detection methods (MSLSTIPT [5], IMNN-LWEC [70], SRSTT [7], 4DST-BTMD [28], STRL-LBCM [71], 4D-TR [29], 4D-TT [29], and NFTDGSTV [101]), deep learning-based SIRST detection methods (ACM [113], ALCNet [46], Res-UNet [33], DNA-Net [30], ISNet [53], UIUNet [31], AGPCNet [114], MSHNet [115], SCTransNet [20], RPCANet [32], and ILNet [116]), and deep learning-based MIRST detection methods (Res-UNet+DTUM [6], STDMA Net [23], Res-UNet+RFR [4], and DQAligner [117]) on the NUDT-MIRSDT dataset, the NUDT-MIRSDT-HiNo dataset, and the IRDST-simulation dataset. All traditional methods were implemented with their default parameters. We adopted 0 as the threshold for

all deep learning-based methods, i.e., 0.5 after sigmoid operation [6], [58]. Quantitative results are presented in Tables 2, 3 and Fig. 11, and qualitative results are shown in Fig. 12.

4.2.1 Quantitative results

As shown in Table 2 and Fig. 11, our DeepPro outperforms the state-of-the-art methods in all scenarios. Especially, on the low-SNR subset and the NUDT-MIRSDT-HiNo dataset, DeepPro on average improves P_d by 3.59% and 7.52%, while reducing F_a by 2.11×10^{-5} and 0.19×10^{-5} , respectively, over sub-optimal methods (i.e., DTUM [6], STDMA Net [23]). The results verify that the temporal profile has significant advantages in containing more distinctive characteristics for IRST detection. That is, long-term temporal information is worthy of attention. Besides, among the deep learning-based methods including the single-frame ones, our DeepPro has the fewest parameters and achieves the second-highest inference efficiency and the second-lowest GFLOPs per frame. That is because, DeepPro performs calculations (e.g., addition and multiplication) only in the time dimension for detection, which still benefits from the superiority of the temporal profile.

From the results of the state-of-the-art methods, we can also observe the advantages of temporal information. In most scenarios (e.g., the NUDT-MIRSDT dataset and the NUDT-MIRSDT-HiNo dataset), deep learning-based SIRST detection methods perform better than traditional MIRST detection methods. However, on the low-SNR subset, some traditional methods (e.g., SRSTT, 4D-TR, 4D-TT, and 4DST-BTMD) significantly outperform deep learning-based SIRST detection methods. That is because, it is hard to detect dim targets (e.g., $SNR \leq 3$) without temporal information. Moreover, SRSTT [7] in traditional methods and STDMA Net [23] in deep learning-based methods achieve much better detection performance than other existing methods on dim targets and complex scenarios (e.g., the NUDT-MIRSDT-HiNo dataset), because they can extract

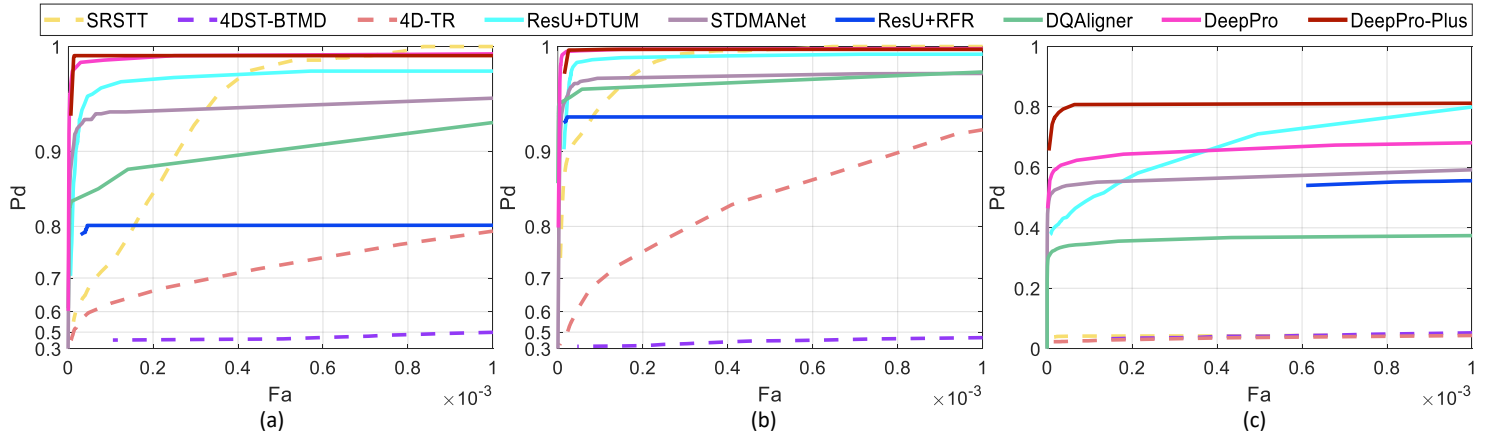


Fig. 11. ROC performances of different methods on the NUDT-MIRSDT dataset and the NUDT-MIRSDT-HiNo dataset. (a) NUDT-MIRSDT ($SNR \leq 3$). (b) NUDT-MIRSDT. (c) NUDT-MIRSDT-HiNo.

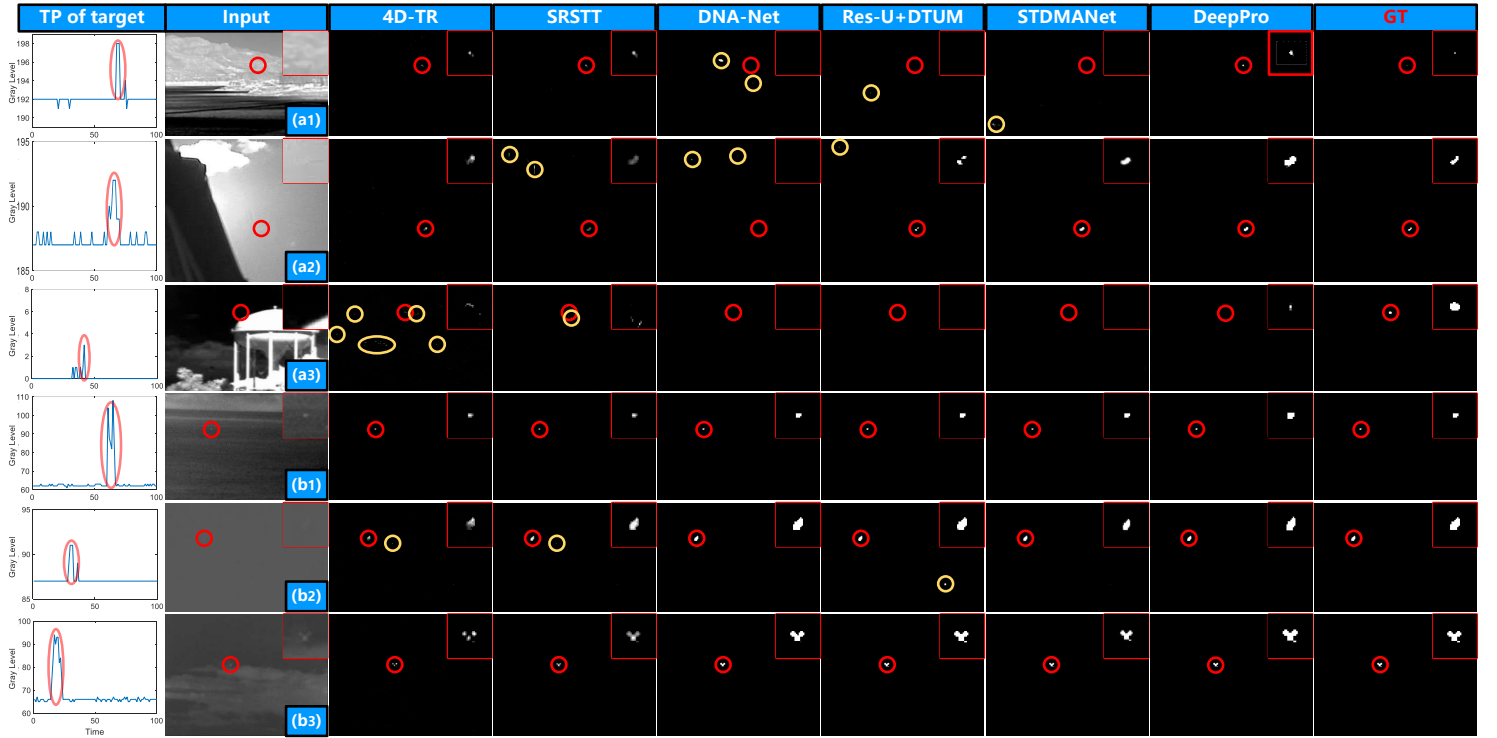


Fig. 12. Visual comparison on the NUDT-MIRSDT dataset. (a1-a3) Results on the NUDT-MIRSDT ($SNR \leq 3$) subset. (b1-b3) Results on the NUDT-MIRSDT ($SNR > 3$) subset. The first column shows the temporal profile (TP) of the target central pixel. For better visualization, the target area is enlarged in the top-right corner and highlighted with a red circle. The false alarm area is marked with a yellow circle.

longer temporal features from more input frames (i.e., 30 and 20, respectively) in a sequence.

4.2.2 Qualitative results

We compared the qualitative results of different detection methods on the NUDT-MIRSDT dataset, as shown in Fig. 12. Compared with the state-of-the-art methods, DeepPro generates better visual results, especially on dim targets (e.g., $SNR \leq 3$) nearly invisible spatially, due to the enhanced signal distinguishability in the temporal profile. In complex scenarios (e.g., a1, a2, and a3), traditional methods and deep learning-based SIRST methods have significant disadvantages in suppressing false alarms, while most deep learning-based MIRST detection methods (e.g., STDMA Net, and our DeepPro) with long sequence as input perform well. That shows the importance of long temporal information for IRST detection.

TABLE 4
Detection results achieved by variants of DeepPro with and without TPro on the NUDT-MIRSDT dataset and the NUDT-MIRSDT-HiNo dataset.

Datasets	Models	P_d	F_a	AUC
NUDT-MIRSDT	DeepPro w/o TPro	97.34	1.96	0.9954
	DeepPro	98.50	0.72	0.9973
$SNR \leq 3$	DeepPro w/o TPro	92.44	3.39	0.9902
	DeepPro	95.84	0.52	0.9952
NUDT-MIRSDT-HiNo	DeepPro w/o TPro	53.21	1.75	0.9137
	DeepPro	59.17	1.76	0.9638

4.3 Model analyses

4.3.1 Effectiveness of TPro

To verify the effectiveness of TPro in IRST detection, we compared the quantitative results of the variant without TPro on the NUDT-MIRSDT dataset and the NUDT-MIRSDT-HiNo

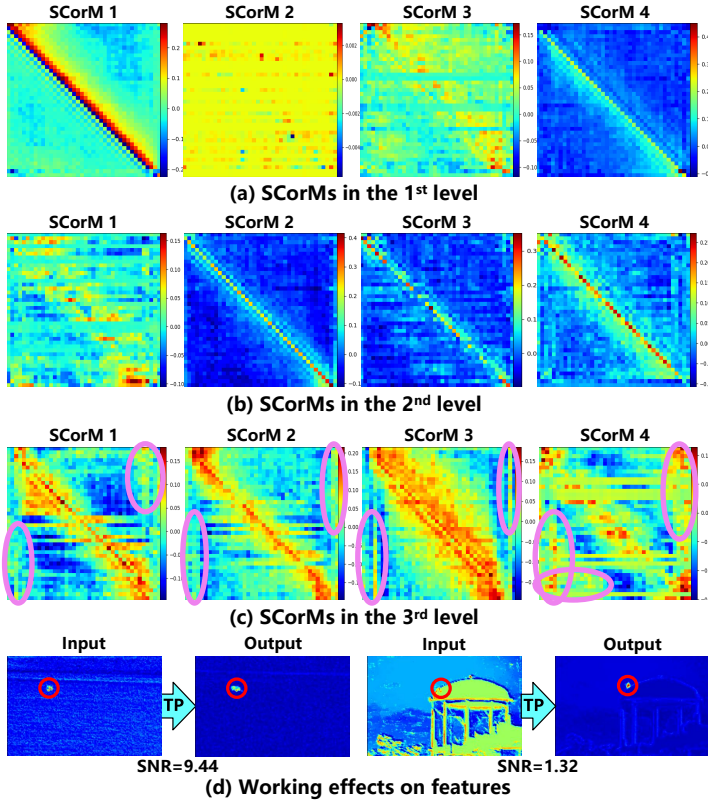


Fig. 13. Visualization of statistic-based correlation matrixes in our DeepPro. (a) SCorMs in the first level which focus on all moments and the target occurrence moments (i.e., the diagonal). (b) SCorMs in the second level which focus on the target occurrence moments. (c) SCorMs in the third level which focus on the moments both proximal and distal to the target presence. The violet circles represent significant moments other than the moments near the target presence. (d) Working effects of TPro on features. For better visualization, the target area is highlighted by a red circle.

dataset. Specifically, we replaced TPro with the same group number (i.e., $m = 4$) of $1 \times 1 \times T$ convolutions. The results are shown in Table 4. It can be observed that DeepPro without TPro suffers 3.4% and 5.96% reductions in terms of P_d on the $SNR \leq 3$ subset and the NUDT-MIRSDT-HiNo dataset, respectively, and suffers a 2.87×10^{-5} increase in terms of F_a on the NUDT-MIRSDT dataset. This demonstrates that TPro plays an important role in extracting sufficient distinctive characteristics from the temporal profile.

To further explore how TPro works, we visualized the statistics-based correlation matrixes (SCorM) and their input and output features in Fig. 13. Different SCorMs contain different statistical properties of the signals. They focus on the intensity variations of signals in different periods and mine different temporal profile features. Fig. 13(a) shows that, in the first level, some SCorMs pay attention to all moments of the input sequence (e.g., SCorMs 2 and 3), and others pay attention to the moments close to the target existence (e.g., SCorMs 1 and 4). In the second level, most SCorMs focus on the target occurrence moment, as shown in Fig. 13(b). In the third level shown in Fig. 13(c), there are some purple circles marking the moments with the locally highest weights except the moments when the targets occur. That is, SCorMs in this level focus on the moments near the target presence and the farthest moments before and after the target presence. That also confirms **Phenomenon 2** in Section 3.2.2, i.e., *distant information is crucial for predictions like recent information*. Different SCorMs work together to extract various temporal profile information and highlight the target

TABLE 5
Detection results achieved by our DeepPro with different lengths of input frames on the NUDT-MIRSDT-HiNo dataset.

T	P_d	F_a	AUC	#Params (M)	GFLOPs	FPS
5	23.14	2.02	0.8339	0.119	0.92	165.53
10	46.44	7.18	0.9634	0.123	0.93	171.99
20	57.49	1.72	0.9726	0.138	0.96	177.94
40	59.17	1.76	0.9638	0.197	1.01	184.55
60	59.63	2.68	0.9797	0.293	1.07	190.52
80	60.21	3.61	0.9804	0.429	1.12	183.47

TABLE 6
Detection results achieved by our DeepPro with different numbers of SCorMs on the NUDT-MIRSDT-HiNo dataset. The best results are in bold.

m	P_d	F_a	#Params (M)	FPS
0	-	-	0.117	-
1	55.12	2.44	0.137	180.76
2	59.11	3.12	0.157	180.40
4	59.17	1.76	0.197	184.55
8	59.17	3.27	0.275	184.21

area. Especially, the dim target is significantly enhanced after TPro or temporal profile calculations as shown in Fig. 13(d).

4.3.2 Length of temporal profile information

To explore the impact of the length of the temporal profile information used for detection, we changed the number T of input frames to 5, 10, 20, 60, and 80, as well as the size of SCorM (i.e., $T \times T$). The results are summarized in Table 5. It can be observed that with the extension of temporal information, P_d and F_a are significantly improved, and the inference efficiency (i.e., FPS) is also gained, albeit with a slight rise in GFLOPs. However, when the length of the time dimension exceeds 40 (e.g., $T = 60, 80$), there is no significant improvement in the detection performance, while the number of parameters and GFLOPs continue to rise gradually. Furthermore, when $T = 5$, DeepPro exhibits significantly degraded detection performance due to the insufficient temporal profile information. This underscores that effective temporal probing relies on long-term temporal information.

To understand why utilizing short temporal profile information performs worse, we visualized the SCorMs in the third level with different lengths in Fig. 14. In predicting the signal property at a given moment, the importance of signals at other moments is only related to their relative time distances rather than their absolute times. Consequently, the SCorMs should be symmetric about the minor diagonals. When the length is shorter than 40, the matrices deviate from this expected symmetry in the third level, e.g., $T = 10$ and $T = 20$ in Fig. 14. This illustrates that it is difficult to learn consistent essential features or accurate statistical properties from short temporal information.

Notably, we compared the results of STDNet [23] with the input of length 20 in Table 2 and the results of the variant with $T = 20$ in Table 5. It can be observed that the detection performance of our DeepPro is obviously superior in all terms of P_d , F_a , AUC, #Params, GFLOPs, and FPS. This demonstrates that *the temporal profile information is more important than local spatial-temporal information in IRST detection*.

4.3.3 Analysis of multiple SCorMs

To investigate the effect of multiple SCorMs on learning variation characteristics in the temporal profile, we changed the number m to 1, 2, and 8. Note that different number

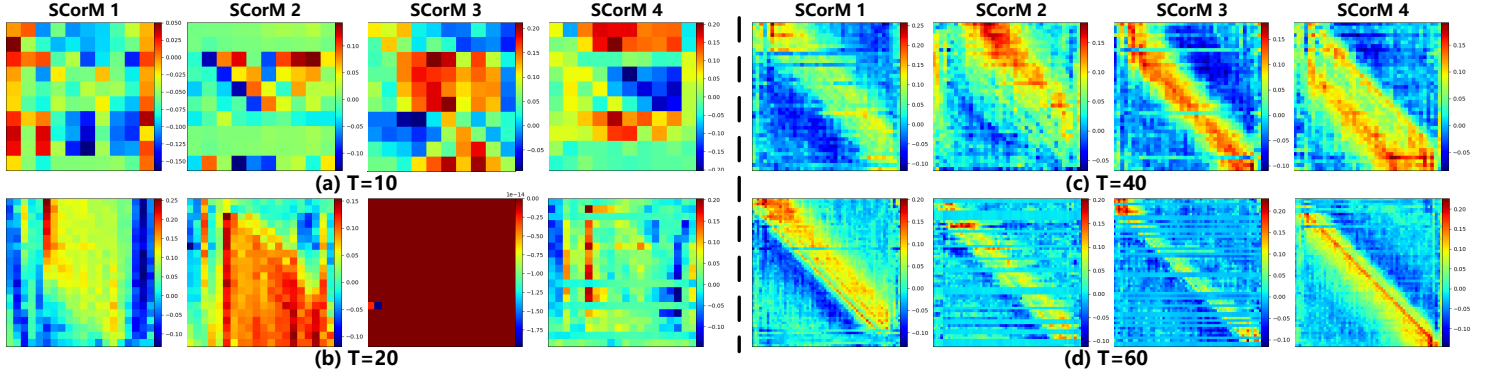


Fig. 14. Visualization of the statistic-based correlation matrixes in the third level with different lengths. (a) SCorMs with length 10 and (b) SCorMs with length 20 are irregular. (c) SCorMs with length 40 and (d) SCorMs with length 60 are approximately symmetric about the minor diagonals. (SCorMs with length 20 are also approximately symmetric about the minor diagonals in the second level as shown in the Appendices. This explains why DeepPro with $T = 20$ demonstrates acceptable performance as compared to the network with $T = 40$, and even performs much better than STDManet.)

TABLE 7

Detection results achieved by several variants of DeepPro with different levels on the NUDT-MIRSDT-HiNo dataset. The best results are in bold.

levels in DeepPro	P_d	F_a	AUC
1 st	53.50	13.82	0.9355
2 nd	54.77	2.11	0.9769
3 rd	56.45	1.87	0.95776
1 st + 2 nd	58.13	5.96	0.9853
2 nd + 3 rd	57.84	1.97	0.9784
1 st + 2 nd + 3 rd	59.17	1.76	0.9638

SCorMs have identical GFLOPs. We compared the detection performances of these variants on the NUDT-MIRSDT-HiNo dataset. The results are reported in Table 6. Compared with a single SCorM, multiple SCorMs significantly improve P_d . In particular, the variant with four SCorMs achieves the best detection performance in terms of P_d and F_a . That is because multiple SCorMs can learn rich correlation properties, which is beneficial for mining temporal profile information, like multi-head attention [52]. However, more SCorMs are not cost-effective, since there is no obvious performance improvement but increases in the number of parameters. In addition, the parameter amount of one SCorM is only 6.56K, and that of other structures in DeepPro is only 0.117M. Our DeepPro is extremely lightweight and effective.

4.3.4 Analysis of the level structure

To explore the effect of the level of the DeepPro structure, we removed some levels shown in Fig. 9, and the comparison results are in Table 7. It can be observed that fewer levels result in higher false alarm rates and lower detection rates, while the multi-level structure significantly improves the detection performance. This is because, with the decrease in the spatial resolution of the features, the process of a target moving through a specified location becomes slow and obvious. Meanwhile, the interference of noise is reduced since most noise is spatially uncorrelated. Consequently, the signal of the target in the temporal profile becomes broader and more distinctive, making it easier for the model to probe the anomaly signals. Furthermore, the multi-level architecture enables the network to capture more comprehensive temporal profile information of the target regions, enriching the feature representation.

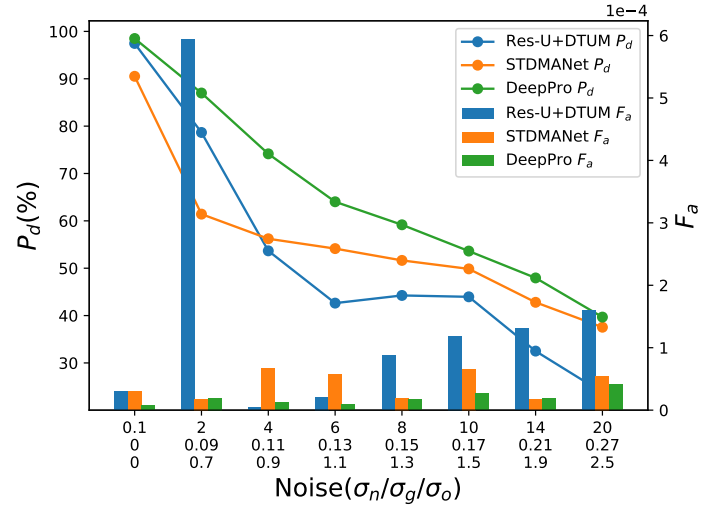


Fig. 15. Results of different networks in scenes under different noise intensity conditions.

4.4 Robustness to noise intensities

We explored the robustness of different networks to the noises with different intensities. We changed the standard deviation σ_n of Gaussian white noise, the distribution range σ_g of the multiplicative noise (related to the non-uniformity), and the standard deviation σ_o of the additive noise to different values. Then, we evaluated the detection performances of different networks on different conditions, as shown in Fig. 15.

As the intensity of the noise increases, the detection performances of all methods experience degradations. Under extreme noise conditions, Res-UNet+DTUM suffers an obvious increase in terms of F_a , whereas our DeepPro consistently maintains a low F_a . In all conditions, our DeepPro achieves significantly superior detection performance. That is because, the temporal probe mechanism can learn more reliable and richer correlation features of different signals in the temporal profile. Besides, the characteristics of long-term temporal variation enable robust discrimination between varying levels of noise, clutter, and target signals. Therefore, our DeepPro is robust to different noise intensity conditions.

TABLE 8

Detection results achieved by different versions of our DeepPro on the NUDT-MIRSDT dataset and the NUDT-MIRSDT-HiNo dataset.

Models	$SNR \leq 3$		NUDT-MIRSDT			NUDT-MIRSDT-HiNo			#Params (M)	GFLOPs	FPS
	P_d	F_a	P_d	F_a	AUC	P_d	F_a	AUC			
DeepPro	95.84	0.52	98.50	0.72	0.9973	59.17	1.76	0.9638	0.197	1.01	184.55
DeepPro-Plus	99.24	1.65	99.71	2.69	0.9978	76.23	1.69	0.9171	0.284	3.89	224.05
DeepPro+S	98.68	0.83	99.60	1.02	0.9976	76.52	1.03	0.9461	0.757	5.22	163.69

4.5 DeepPro-Plus: Combination with little spatial information

As aforementioned, we confirm that temporal profile information is very important for IRST detection. Only leveraging the temporal profile information can achieve superior performance as compared with the state-of-the-art methods that use spatial or local spatial-temporal information. Then, a question arises naturally: How about combining temporal profile information with little spatial information? To answer it, we further developed another version of our DeepPro, called DeepPro-Plus, which replaces all temporal convolutions in temporal feature extraction with spatial-temporal convolution and retains only the first level in DeepPro. All TD-ResBlocks in DeepPro are changed into spatial temporal difference residual blocks (STD-ResBlock). The details are described in the Appendices.

4.5.1 Comparison to the state-of-the-arts

We compared our DeepPro-Plus with the same state-of-the-art methods in Subsection 4.2. The quantitative results achieved by DeepPro-Plus in different scenes and on different targets are shown in Table 8. Furthermore, we expanded our evaluation on a more diverse, large-scale dataset (the IRSatVideo-LEO dataset [4]) with smaller targets for a thorough comparison. See the Appendices for further results.

In all scenarios, DeepPro-Plus achieves the highest detection performance with absolute superiority in almost all aspects as compared with the state-of-the-art methods and DeepPro in Table 2. Especially in scenes with strong noise, DeepPro with little spatial calculations improves P_d by more than 20%, while maintaining a lower F_a . Although GFLOPs and the number of parameters are increased, the increment is negligible compared to other existing methods. That is because, the spatial receptive field of DeepPro-Plus is only 19×19 in theory. Besides, the real-time performance (i.e., FPS) is improved because of less-level structures. Therefore, it can be concluded that combining temporal profile information with little spatial information is further beneficial for IRST detection. In addition, the temporal profile information is more important.

4.5.2 Ablation study

In this part, we validate that one level is enough when combining little spatial information, and a deep spatial-temporal feature extraction structure is unadvisable design for IRST detection.

1) Number of levels: From DeepPro to DeepPro-Plus, we maintained only one level in the structure. To explain its effectiveness, we compared DeepPro-Plus with DeepPro+S obtained by only replacing all temporal convolutions with spatial-temporal convolutions in DeepPro. The results are shown in Table 8. We find that the performance gap in IRST detection between DeepPro+S and DeepPro-Plus is minimal, but the structure with three levels (i.e., DeepPro+S) requires substantially more computational overhead. Consequently, little spatial information in one level is enough to help the network to

TABLE 9

Detection results achieved by several variants of DeepPro-Plus with different numbers of STD-Resblocks on the NUDT-MIRSDT-HiNo dataset. The best results are in bold.

n	P_d	F_a	AUC	#Params (M)	GFLOPs
2	76.23	1.69	0.9171	0.284	3.89
3	73.05	2.26	0.9100	0.515	7.68
4	72.76	3.46	0.9114	0.745	11.47

TABLE 10

Detection results achieved by different state-of-the-art methods on the real-world RGBT-Tiny dataset.

	Methods	P_d	F_a	AUC
SF	AGPCNet [114]	51.05	10436.86	0.7296
	MSHNet [115]	73.45	14.37	0.9405
	SCTransNet [20]	76.13	6.35	0.8852
	RPCANet [32]	1.50	0.03	0.4824
	ILNet [116]	72.72	16.57	0.9407
MF	Res-U+DTUM [6]	68.81	4.79	0.8462
	STDManet [23]	45.76	4.61	0.8865
	Res-U+RFR [4]	59.42	3.48	0.8155
	DQAligner [117]	58.08	97.57	0.8264
	DeepPro-Plus	77.83	4.47	0.9730

further capture richer temporal profile information of the target region. This reveals a notion that extensive spatial receptive field and complex feature extraction are unessential, but focusing on the temporal profile is crucial.

2) Depth of spatial-temporal feature extraction: To explore the effect of the depth of the spatial-temporal feature extraction structure, we adjusted the number of STD-ResBlocks n to 2, 3, and 4. The results are shown in Table 9. It can be observed that when $n = 2$, the network performs best with a 3.18% P_d increase and a 0.57×10^{-5} F_a decrease as compared with the second best results. This reveals that the deeper structure (e.g., more STD-ResBlocks) or the larger spatial receptive field does not bring a performance improvement in IRST detection, but an efficiency decline. These results can also explain Phenomenon 1 that the influential pixels are concentrated on the temporal profile of the target since the temporal profile information is more important.

4.6 Performance in the real world

We tested the detection performance of our method on real scenarios from the RGBT-Tiny dataset [107]. We selected 45 and 15 scenarios containing IRSTs from its training set and test set, respectively. In the dataset, there are various small targets (e.g., ships, pedestrians, cars, drones, and planes) with different moving speeds, and diverse scenarios with dynamic backgrounds and camera movements. For training, we generated rough Gaussian masks as labels according to the bounding boxes provided by its authors. The quantitative results are shown in Table 10, and the qualitative results are shown in Fig. 16.

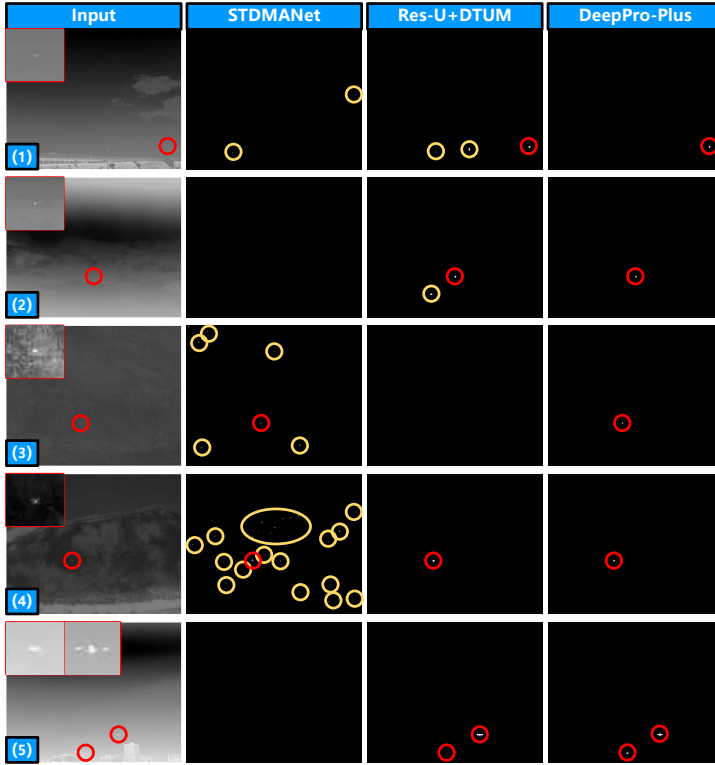


Fig. 16. Visual comparison on the real-world RGBT-Tiny dataset. For better visualization, the target area is enlarged in the top-left corner and highlighted with a red circle. The false alarm area is marked with a yellow circle.

From Table 10, it can be found that the P_d of DeepPro-Plus is 9.02% higher than that of the sub-optimal multi-frame method (i.e., DTUM) in the case of close F_a . This shows that our method can achieve state-of-the-art detection performance in the real world. From Fig. 16, we can observe that DeepPro-Plus achieves good detection performance in various scenes. In scene (3), the background and target are very dim and the noise is relatively prominent. Other methods produce more false alarms or miss detection, while our method can still maintain good performance. As for scene (4), the background is changed due to camera movement. STDMArNet produces a lot of false alarms, while our method and DTUM can still adapt and perform well.

4.7 Failure cases

When testing our method in real-world scenarios, there are some failure cases of miss detection, as shown in Fig. 17.

Miss detecting obscured targets. The target (m1) in Fig. 17 is obscured by trees, leading to an incomplete signal in the temporal profile. The target signal is so narrow as a spike. This target is missed, and the reason can be twofold. On the one hand, this may be due to the incomplete signal. On the other hand, it may be due to the lack of such samples in the training set.

Miss detecting stationary targets. The targets (m2), (m3), and (m4) are stationary during a period of time. From their temporal profiles, it can be observed that there is no regular temporal profile information like that of the moving targets. For a long time (over 40 frames), the signal is maintained at a fixed gray level. It is hard to detect stationary objects from the temporal profile. Therefore, our method is targeted at moving IRSTs.

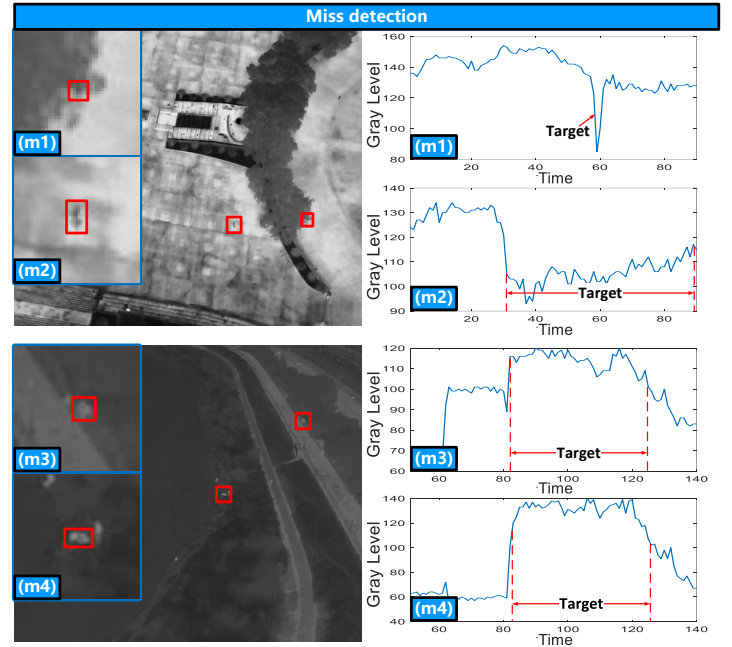


Fig. 17. Failure cases in the real world and their gray variations in the temporal profile. (m1) is a pedestrian obscured by trees. (m2), (m3), and (m4) are stationary pedestrians. Their signals in the temporal profile are different from the ordinary moving target signals as shown in Figs. 3, 4 and 5.

5 CONCLUSION

In this paper, we consider the essential domain and information for precise, robust, and efficient infrared small target (IRST) detection. We observe the targets from the temporal profile. We reveal some overlooked but essential temporal profile information, that is, the global temporal saliency of the target and the correlation information between target signal and other signals. We built the first prediction attribution tool, and validated the robustness of the temporal profile information (long-term temporal information) by applying it to IRST detection networks. Based on our research, we remodel the IRST detection task as a one-dimensional signal anomaly detection task and propose an efficient and effective IRST detection network, deep temporal probe network (DeepPro), in which the calculations (i.e., multiplication and addition) are only in the time dimension. Numerous experiments on widely-used benchmarks were conducted to confirm the effectiveness of our methods. It is exciting that our DeepPro outperforms the state-of-the-art methods especially on dim targets and in complex scenarios with extremely few parameters and extremely high processing efficiency. Through this work, we reveal the paramount importance of temporal profile information.

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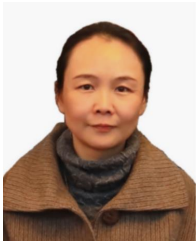
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